

ACE 535: Data Science for Applied Economics

Spring 2026

Class Meets: TR, 3:30 - 5:10 PM
Instructor: Professor Jared Hutchins
Contact: jhtchns2@illinois.edu
Graduate Credit Hours: 4

316N Mumford
Office: 312 Mumford
Office Hours: TR, 2:15-3:15pm
(or by appointment)

Course Description

The ability to obtain, process, and analyze data using coding and algorithms has become an absolutely essential skill for anybody engaged in economic analysis. In the digital age, there is more data available than ever before on human behavior: from analyzing an elected official's opinion from Twitter to identifying a farmer's crop choices through satellite images. For those engaged in academic research in economics, these new data sources have drastically expanded the number of research questions that can be answered and revolutionized how we answer classic questions in the field. For those working outside of academia, the ability to process and analyze large datasets to provide insight has become an essential skill for producing useful analysis on a problem or question.

The goal of this course is to teach students in applied economics how to use data science tools and workflow for answering questions in economics. Using the python programming language and git version control, we will cover obtaining data via scraping and APIs, processing and cleaning data using python, and analyzing data via data visualization and basic machine learning techniques. The course will broadly cover the basics of text, spatial, and numeric data with an emphasis on their uses in analyzing economic questions and conducting research.

Data science can be broadly defined as “an approach to data analysis with a foundation in code and algorithms.” The main goals of data science include drawing useful conclusions from large and diverse datasets through exploration, prediction, and inference. Data science is fundamentally different than most previous approaches to data analysis because:

- It uses **diverse data sources**, such as text, image, spatial, or numeric.
- There is an emphasis on **work flow** that uses coding and version control to maximize reproducibility and transparency.
- It uses **new tools and approaches** for answering questions about data.

Course Objectives

In light of this, by the end of the course we will be able to:

1. Obtain and process text, image, and numeric data using Python.
2. Analyze data using basic data visualization and machine learning in Python.
3. Construct a git repository and collaborate on a research project on Github.
4. Document code and communicate results using Jupyter notebooks.

To achieve these objectives, we will learn three main tools:

- **Python**, an object-oriented and general purpose programming language that can be used for reading, processing, and analyzing data.
- **Git**, a version control software for documenting and collaboration, as well as its online platform Github.
- **Jupyter notebooks**, a development environment for interactive programming that supports markdown.

Course Delivery

We will meet **in-person** in Mumford 316N on Tuesdays and Thursdays 4:00 - 5:20 pm. For those interested in recorded lectures to review, there is a catalog of two years of previous lectures than can be made available through MediaSpace. Office hours for the instructor can be in person in Mumford 312 or over Zoom. The Zoom link for office hours can be found in the Syllabus Appendix.

Instead of a learning management system like Canvas, I use **Slack** to communicate with students about the course and **GitHub** to deliver course materials. The assignments are submitted and grading using **GitHub Classroom**. The data for completing in-class examples and the assignments is stored on a **Box** folder for use by the class. The link to the Box folder, GitHub repository, and Slack group are made available in the Syllabus Appendix.

Prerequisites

There is no formal prerequisite for this course, but to be the most successful in this course the student *should be at least a masters student, should have a baseline knowledge of econometrics and/or statistics, and should have some experience coding (e.g. STATA, SAS, Matlab)*, though experience coding Python is not assumed for this course.

Since much of this course is learning how to write code, students should note that **to learn to code, the best way is to do it. A lot.** Part of learning how to code will also be learning what resources exist to help you overcome problems, so while the lectures and homeworks exist as guides **how much you learn is a function of how much effort you put into practicing and supplementing your existing knowledge.**

Grading and Assignments

Grading will be based on two components:

- **Five assignments** due throughout the semester (60 points total).
 - Submissions must be in the form of **Jupyter Notebook**, submitted to GitHub Classroom (links in Syllabus Appendix).
 - Group work is encouraged, but write-ups **must be individual**.
 - There is one additional homework at the end of the semester, Homework 4, which is 10 extra credit points.
- One **final analysis project and presentation**, done in groups (15 points GitHub Repo + 15 points write-up + 10 points presentation = 40 points).
 - The project must answer an applied economics question using data.
 - Groups must be formed by **Feb 13**, and the topic has to be approved by me by **March 27**.
 - Submission will be 1) a presentation done at the end of the semester, 2) a Github repository for your group's project, and 3) a write up of the findings of your project.

Grading Rubric

Assignment	Points	Due Date
Homework 0	5	February 12
Homework 1	15	March 12
Homework 2	20	April 7
Homework 3	20	April 30
Final Presentation	10	May 12 - May 14
Final Project	30	May 14
Total Points	100	
Homework 4 (Extra Credit)	10	May 14

Grades will be assigned as follows:

- A = 85-100%
- B = 80-85%
- C = 70-79%
- D = 60-69%
- F = 0-59%

Resources

There is no textbook for this course, as most of the learning in this course is self-guided to some extent. As you come across problems in solving assignments and the like, these resources can help you overcome issues.

Big Picture Resources

- [Learn Python](#) (good for basics)
- [Python for Data Science](#) (most useful one in my opinion)
- [Coding for Data course](#)
- [Inferential Thinking](#), a Berkeley course on data science:
- [Fundamentals of Data Visualization](#)
- [Python Graph Gallery](#)

Specific Resources

- Stack Overflow for very specific, coding problems.
- [DataCamp Cheat Sheets](#), for git, Jupyter, and various python operations.

Absence Policy

There is no penalty for absences in the class, but it is expected that students regularly show up to participate and be active participants in class. Students who are occasionally not able to attend for reasons of sickness or other special situations are encouraged to talk to the instructor and make use of lecture materials that are being posted to keep pace with the class.

Academic Integrity

The University of Illinois Urbana-Champaign [Student Code](#) should also be considered as a part of this syllabus. Students should pay particular attention to Article 1, Part 4: Academic Integrity.

Academic dishonesty may result in a failing grade. Every student is expected to review and abide by the [Academic Integrity Policy](#). Ignorance is not an excuse for any academic dishonesty. It is your responsibility to read this policy to avoid any misunderstanding. Do not hesitate to ask the instructor(s) if you are ever in doubt about what constitutes plagiarism, cheating, or any other breach of academic integrity.

Family Educational Rights and Privacy Act (FERPA)

The federal Family Educational Rights and Privacy Act ([FERPA](#)) affords students certain rights with respect to their education records.

Mental Health

Significant stress, mood changes, excessive worry, substance/alcohol misuse or interferences in eating or sleep can have an impact on academic performance, social development, and emotional wellbeing. The University of Illinois offers a variety of confidential services including individual and group counseling, crisis intervention, psychiatric services, and specialized screenings which are covered through the Student Health Fee. If you or someone you know experiences any of the above mental health concerns, it is strongly encouraged to contact or visit any of the University's resources provided below. Getting help is a smart and courageous thing to do for yourself and for those who care about you.

- Counseling Center (217) 333-3704
- McKinley Health Center (217) 333-2700
- National Suicide Prevention Lifeline (800) 273-8255
- Rosecrance Crisis Line (217) 359-4141 (available 24/7, 365 days a year)

If you are in immediate danger, call 911.

Community of Care

As members of the Illinois community, we each have a responsibility to express care and concern for one another. If you come across a classmate whose behavior concerns you, whether in regard to their well-being or yours, we encourage you to refer this behavior to the Connie Frank CARE Center (formerly the Student Assistance Center) in the Office of the Dean of Students. You may do so by calling 217-333-0050 or by submitting an [online](#)

[referral](#). Based on your report, staff in the Student Assistance Center will reach out to offer support and assistance.

Further, as a Community of Care, we want to support you in your overall wellness. We know that students sometimes face challenges that can impact academic performance (examples include mental health concerns, food insecurity, homelessness, personal emergencies). Should you find that you are managing such a challenge and that it is interfering with your coursework, you are encouraged to contact the [Connie Frank CARE Center](#) (formerly the Student Assistance Center) in the Office of the Dean of Students for support and referrals to campus and/or community resources.

Students with Disabilities

The University of Illinois is committed to ensuring that all students, including those with disabilities, do not experience barriers to learning and participating fully in class. If you have a letter of accommodation from DRES and have not already given it to me, please do so as soon as possible to ensure your accommodation needs are met. To obtain disability-related academic adjustments and/or auxiliary aids, students with disabilities must contact Disability Resources and Educational Services (DRES) as soon as possible. To contact DRES, you may:

To contact DRES, you may visit 1207 S. Oak St., Champaign, call 217-333-1970, e-mail disability@illinois.edu or visit the DRES website at <http://www.disability.illinois.edu/>. Here is the direct link to apply for services at DRES, <https://www.disability.illinois.edu/applying-services>.

Disruptive Behavior

Behavior that persistently or grossly interferes with classroom activities is considered disruptive behavior and may be subject to disciplinary action. Such behavior inhibits other students' ability to learn and an instructor's ability to teach. A student responsible for disruptive behavior may be required to leave class pending discussion and resolution of the problem and may be reported to the [Office for Student Conflict Resolution](#). For Disciplinary action visit:

- Email: conflictresolution@illinois.edu
- Contact: 217-333-3680

Emergency Response Recommendations

Emergency response recommendations and campus building floor plans can be found on the [Division of Public Safety website](#). I encourage you to review this website within the first 10 days of class.

Religious Observances

It is the policy of the University of Illinois Urbana-Champaign to reasonably accommodate its students' religious beliefs, observances, and practices that conflict with a student's class attendance or participation in a scheduled examination or work requirement, consistent with state and federal law. Students should make requests for accommodation in advance of the conflict to allow time for both consideration of the request and alternate procedures to be prepared. Requests should be directed to the instructor. The [Office of the Dean of Students](#) provides an optional resource to assist students in making such requests.

Sexual Misconduct Reporting Obligation

The University of Illinois is committed to combating sex-based misconduct. Faculty and staff members are required to report any instances of sex-based misconduct to the University's Title IX Office. In turn, an individual with the Title IX Office will provide information about rights and options, including accommodations, support services, the campus disciplinary process, and law enforcement options.

A list of the designated University employees who, as counselors, confidential advisors, and medical professionals, do not have this reporting responsibility and can maintain confidentiality, can be found here: wecare.illinois.edu/resources/students/confidential.

Other information about resources and reporting is available here: wecare.illinois.edu.

Veterans and Military Students

As a military-friendly institution, and per federal regulations and Illinois statutes, the University of Illinois Urbana-Champaign has established policies and procedures to accommodate military-connected students. In addition to the support available at the Chez Veterans Center, members of the National Guard or Reserves and active-duty military personnel with military obligations (e.g., deployments, training, drills) are encouraged to communicate these, in advance whenever possible, to the instructor. The policy for Excused Absences and Departure from the University for U.S. Military or other U.S. National Defense Services can be found at [Student Code website](#).

Course Schedule

Subject to change given needs of the semester.

Week	Module	Learning Objectives/ Week Topics
	Introduction	The basics of Git, Jupyter, and the Python packages pandas, numpy, matplotlib, and requests.
1		- Python, git, and Jupyter basics
2		- Pandas, numpy, matplotlib
3		- requests, API basics
4		- Homework 0
	Text as Data	How to clean and process text for natural language processing (NLP) such as n-grams and sentiment analysis.
5		- Python processing text
6		- HTML parsing and scraping, basic NLP
7		- Homework 1
	Images as Data	How to clean and process images for basic GIS analysis such as reprojections and zonal statistics.
8		- Image editing, spatial data types
9		- Spatial statistics, basic mapping
10		- Homework 2
	Numbers as Data	How to do effective data processing with vectorizing, parallelizing, and chunking.
11		- Advanced pandas and numpy operations
12		- Distributed programming, scaling and parallelizing;
13		- Homework 3
	Data Analysis	Advanced data visualization, supervised and unsupervised machine learning.
14		- Advanced visualization
15		- Unsupervised learning, feature generation
16		- Supervised learning, cross-validation
17	Presentations	